$v = 6.01\ \mathrm{m\,s^{-1}}$

7 **(a)** $f = \frac{1}{T} = \frac{1}{20 \times 10^{-3}} = 50\ \text{Hz}$

(b) $V_{XY,rms} = \frac{17.0}{\sqrt{2}} = 12.0\ \text{V}$

(c)

$$\frac{N_P}{N_S} = \frac{V_P}{V_S}$$

$$\frac{4000}{200} = \frac{V_P}{12.02}$$

$$V_P = 240\ \text{V}$$

(d) There is a potential difference across the resistor for only half of each cycle.

$$\text{mean power} = \frac{1}{2} \times \frac{{V_{rms}}^2}{R}$$

$$= \frac{1}{2} \times \frac{12.02^2}{4.8}$$

$$= 15.1\ \text{W}$$

(e) For first half of a cycle, from <u>5 ms to 15 ms</u>, the diode is <u>forward biased</u> and <u>the p.d. across the load resistor is the same as V_{XY}</u>.

For the second half of a cycle, from 15 ms to 25 ms, the diode is reverse biased and the p.d. across the load resistor is zero.

8 (a) $T = 4.70 - 1.20 = 3.50$ days